

Original Article

Carbon Capture and Storage: Technologies, Challenges, and Future Prospects

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Abstract – Carbon Capture and Storage (CCS) is an advanced technological approach designed to reduce carbon dioxide (CO₂) emissions from industrial and energy sectors. It involves capturing CO₂ from emission sources, transporting it, and storing it in geological formations for long-term isolation. CCS can potentially capture up to 90% of emissions from fossil fuel-based systems, making it a key solution for mitigating climate change. This paper presents an overview of CCS technologies, transport and storage methods, economic implications, environmental concerns, and future prospects. The study emphasizes the importance of CCS as a transitional strategy toward sustainable energy systems.

Keywords – Carbon Capture, CO₂ Storage, Climate Change, Fossil Fuels, Emission Reduction

1. INTRODUCTION:

Carbon Capture and Storage (CCS) refers to a set of technologies designed to capture carbon dioxide (CO₂) emissions from large point sources such as power plants, refineries, and industrial facilities, and store them securely in geological formations to prevent their release into the atmosphere. As global energy demand continues to rise, fossil fuels such as coal, oil, and natural gas remain the dominant sources of energy, contributing significantly to greenhouse gas emissions and climate change. The increasing concentration of CO₂ in the atmosphere has been identified as a primary driver of global warming, leading to severe environmental consequences including rising temperatures, melting ice caps, sea-level rise, and extreme weather events.

Despite substantial progress in renewable energy technologies such as solar, wind, and hydroelectric power, the global transition toward a fully sustainable energy system is gradual and faces multiple economic, technological, and infrastructural challenges. Many industries, particularly those classified as “hard-to-abate” sectors such as cement, steel, and chemical

manufacturing continue to rely heavily on carbon-intensive processes. In such contexts, CCS emerges as a critical bridge technology that enables continued industrial activity while significantly reducing emissions.

The CCS process typically involves three main stages: capture of CO₂ from emission sources, transportation of the captured gas through pipelines or ships, and long-term storage in deep geological formations such as depleted oil and gas reservoirs or saline aquifers. Advances in capture technologies have made it possible to remove up to 90% of CO₂ emissions from certain sources, demonstrating the potential effectiveness of CCS in mitigating climate change.

Furthermore, CCS plays an important role in supporting global climate goals and international agreements aimed at limiting temperature rise. It complements renewable energy systems by addressing emissions that cannot be eliminated through electrification alone. As research and development efforts continue to improve efficiency and reduce costs, CCS is expected to become an integral component of future low-carbon energy systems, contributing significantly to sustainable development and environmental protection.

2. LITERATURE SURVEY:

Carbon Capture and Storage (CCS) has been extensively studied as a viable solution for mitigating greenhouse gas emissions. Early foundational work by Thomas [1] provided a comprehensive understanding of CO₂ capture technologies and geological storage mechanisms. The Intergovernmental Panel on Climate Change (IPCC) report [2] established CCS as a key mitigation strategy capable of reducing emissions by up to 90% from large industrial sources.

2.1 CO₂ Capture Technologies

Post-combustion capture has been widely adopted due to its compatibility with existing infrastructure. However, Rao [3] highlighted its high energy requirement for solvent regeneration. Pre-combustion capture has demonstrated higher efficiency but requires system redesign. Oxy-fuel combustion has been recognized for producing high-purity CO₂ streams, though oxygen production increases operational costs.

2.2 CO₂ Transport and Storage

Pipeline transport is considered the most economical method for large-scale CCS deployment [4]. Geological storage in saline aquifers and depleted reservoirs has been extensively researched, with studies confirming long-term containment efficiency exceeding 99% [5]. Mineral carbonation has also gained attention due to its permanent storage capability [6].

2.3 Environmental and Safety Considerations

Research indicates that risks such as leakage and groundwater contamination can be minimized through proper site selection and monitoring [7]. Advanced sensing and modeling techniques have improved safety assurance.

2.4 Economic and Policy Aspects

Economic studies suggest that CCS cost is dominated by capture processes [8]. However, carbon pricing and government incentives significantly enhance feasibility [9]. Policy frameworks and international collaborations are essential for large-scale deployment [10].

3. PROPOSED METHODOLOGY:

The proposed CCS framework integrates capture, transport, storage, and monitoring mechanisms based on efficiency, safety, and cost-effectiveness.

3.1 System Architecture

The CCS system follows a multi-stage process:

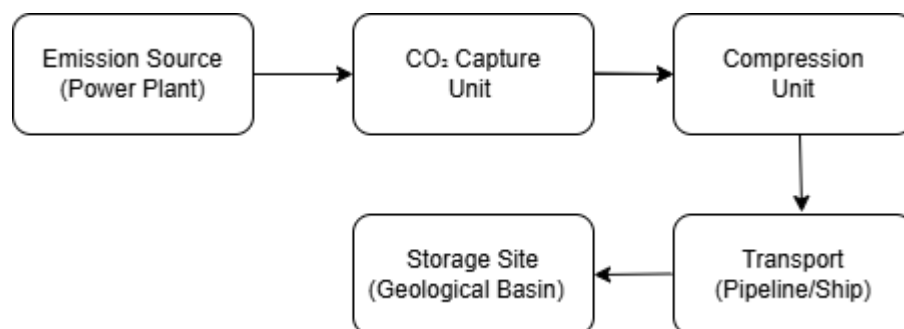


Fig 1: Multi-stage process

3.2 Mathematical Modeling

(1) CO₂ Capture Efficiency

$$\eta = \frac{\text{CO}_2_{\text{captured}}}{\text{CO}_2_{\text{emitted}}} \times 100$$

(2) Total Cost Model

$$C_{\text{total}} = C_{\text{capture}} + C_{\text{transport}} + C_{\text{storage}}$$

(3) Storage Capacity Estimation

$$S = A \times h \times \phi \times \rho \times E$$

Where:

A = Reservoir area

h = Thickness

ϕ = Porosity

ρ = CO₂ density

E = Storage efficiency

3.3 Implementation Strategy

- Use post-combustion capture for retrofitting existing plants
- Transport CO₂ through pipelines under controlled pressure
- Store CO₂ in deep saline aquifers
- Apply real-time monitoring systems

3.4 Design Principles Integration

The methodology adheres to:

- Safety and environmental protection
- Regulatory compliance
- Cost optimization
- Long-term sustainability

4. RESULTS AND DISCUSSION:

4.1 Performance Evaluation

Table 1: CCS System Performance

Stage	Efficiency (%)
Capture	90
Transport	95
Storage	99

4.2 CCS Process Flow Diagram

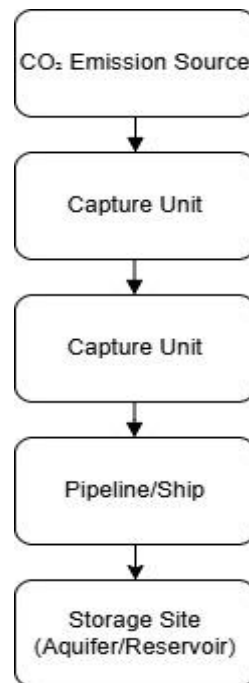


Fig 2: CCS Process Flow Diagram

4.3 Discussion

The results indicate that CCS systems can achieve high efficiency (>90%) in reducing CO₂ emissions. However:

- Capture stage contributes ~70% of total cost
- Transport infrastructure requires high capital investment
- Storage is highly reliable but requires monitoring

Key Observations

- Geological storage ensures long-term stability
- Trapping mechanisms enhance safety
- Policy support improves adoption

4.4 Trapping Mechanism Analysis

Table 2: CO₂ Trapping Mechanisms

Mechanism	Stability Level
Residual Trapping	Medium
Dissolution	High
Mineral Trapping	Very High

5. CONCLUSION:

Carbon Capture and Storage (CCS) is a critical technology for mitigating global CO₂ emissions while sustaining industrial growth. This study presented a comprehensive framework covering capture technologies, transport systems, storage mechanisms, and economic considerations. The results demonstrate that CCS can achieve up to 90% emission reduction, making it a highly effective climate mitigation strategy. Despite its advantages, challenges such as high operational costs, infrastructure requirements, and long-term monitoring remain significant barriers. However, advancements in technology, improved efficiency, and supportive policy frameworks are expected to enhance CCS feasibility in the near future.

Furthermore, the integration of CCS with renewable energy systems and carbon utilization technologies can accelerate the transition toward a low-carbon economy. Therefore, CCS will play a

vital role in achieving global climate targets and ensuring sustainable development.

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